Literature Review Notes: Extreme Events and Realized Volatility Forecasting: Information Augmentation in Linear and Nonlinear Models

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1 Intention Statement

Project Title: *Extreme Events and Realized Volatility Forecasting: Information Augmentation in Linear and Nonlinear Models*

Purpose of the Project

The purpose of this project is to investigate whether information extracted from extreme intraday market events can enhance the prediction of realized volatility in cryptocurrency markets. Traditional forecasting frameworks such as the Heterogeneous Autoregressive (HAR) model rely primarily on aggregated volatility measures across daily, weekly, and monthly horizons. While this structure captures long-memory behavior in volatility, it may fail to incorporate information embedded in high-frequency extreme market movements.

This project explores whether explicitly incorporating features that characterize extreme intraday events can improve volatility forecasting performance and whether nonlinear models are better suited to exploit this additional information.

Motivation

Cryptocurrency markets are characterized by frequent abrupt price movements, liquidity shocks, and clustering of extreme events. These phenomena suggest the presence of event-driven volatility dynamics that may not be fully captured by linear models relying solely on aggregated realized volatility measures.

Recent advances in machine learning offer flexible frameworks capable of modeling nonlinear dependencies and complex temporal structures. However, improvements from such models are often attributed to their higher flexibility rather than their ability to effectively utilize new information.

To address this concern, the project adopts an information-centric perspective: both linear and nonlinear models will be provided with the same set of extreme-event features in order to evaluate whether predictive improvements arise from the information itself or from the model's capacity to exploit it.

Research Question

Can the detection of extreme events in intraday data improve the prediction of realized volatility in cryptocurrency markets?

Hypotheses

H1. Extreme-event indicators extracted from intraday data contain additional predictive information for future realized volatility.

H2. Augmenting traditional volatility forecasting models with extreme-event features improves out-of-sample forecasting accuracy relative to standard HAR specifications.

H3. Nonlinear models, such as Long Short-Term Memory (LSTM) networks, are able to exploit extreme-event information more effectively than linear models.

Expected Contribution

The expected contribution of this project is twofold. First, it evaluates whether event-based information extracted from high-frequency data provides incremental predictive power for realized

volatility forecasting in cryptocurrency markets. Second, it establishes an empirical foundation for the development of event-driven volatility models, including future architectures that integrate point-process dynamics such as Hawkes processes with deep learning frameworks.

By focusing on the role of information rather than solely on model complexity, this project aims to clarify whether improvements in volatility prediction arise from richer data representations or from the flexibility of nonlinear forecasting models.

2 Paper Review

2.1 Alan G. Hawkes (1971) – Spectra of Some Self-Exciting and Mutually Exciting Point Processes

Reference: [18] **Tags:** Keywords **Link:** http://www.jstor.org/stable/2334319

Foundation of a self-excitement point process called Hawkes.

2.2 Yosihiko Ogata (1988) – Statistical Models for Earthquake Occurrences and Residual Analysis for Point Processes

Reference: [28] **Tags:** None **Link:** http://www.jstor.org/stable/2288914

Relevant information to include.

Foundation for likelihood-based estimation used later in finance Hawkes calibrations

2.3 Francis X. Diebold and Roberto S. Mariano (1995) – Comparing Predictive Accuracy

Reference: [13] **Tags:** Keywords **Link:** http://www.jstor.org/stable/1392185

Relevant information to include.

DM test.

Provides tests for equal predictive accuracy across competing forecasts under flexible loss and serial correlation.

2.4 Hochreiter, Sepp and Schmidhuber, Jürgen (1997) – Long Short-Term Memory

Reference: [20] **Tags:** Keywords **Link:** 10.1162/neco.1997.9.8.1735

Relevant information to include.

LSTM foundation.

2.5 Ulrich A. Müller and Michel M. Dacorogna and Rakhal D. Davé and Richard B. Olsen and Olivier V. Pictet and Jacob E. von Weizsäcker (1997) – Volatilities of different time resolutions — Analyzing the dynamics of market components

Reference: [27] **Tags:** Heterogeneous markets, High frequency data **Link:** https://www.sciencedirect.com/science/article/pii/S0927539897000078

Relevant information to include.
heterogeneous market hypothesis.

2.6 Torben G. Andersen and Tim Bollerslev (1998) – Answering the Skeptics: Yes, Standard Volatility Models do Provide Accurate Forecasts

Reference: [1] **Tags:** Keywords **Link:** http://www.jstor.org/stable/2527343

Relevant information to include.
Milestones of RV formulation and properties.

2.7 Torben G. Andersen and Tim Bollerslev and Francis X. Diebold and Heiko Ebens (2001) – The distribution of realized stock return volatility

Reference: [2] **Tags:** Keywords **Link:** https://www.sciencedirect.com/science/article/pii/S0304405X01000551

Relevant information to include.
Evidence of right-skewness of RV. Motivate to use sqrt or log transformation for more gaussian form.

2.8 Andersen, Torben G. and Bollerslev, Tim and Diebold, Francis X. and Labys, Paul (2003) – Modeling and Forecasting Realized Volatility

Reference: [3] **Tags:** continuous–time methods, quadratic variation, realized volatility, high–frequency data, long memory, volatility forecasting, density forecasting, risk management **Link:** https://onlinelibrary.wiley.com/doi/abs/10.1111/1468-0262.00418

Problem statement and contribution

- Problem: Current models fails to capture multi-asset and intraday features in daily forecast of realized volatility and VaR due to complexity and/or curse of dimensionality.
- Main contribution: "We provide a framework for integration of high–frequency intraday data into the measurement, modeling, and forecasting of daily and lower frequency return volatilies and return distributions."
- Why it matters: Model and forecast daily and lower frequency risk measures is important.

Commentaries

"The theory of quadratic variation reveals that, under suitable conditions, realized volatility is not only an unbiased ex-post estimator of daily return volatility, but also asymptotically free of measurement error, as discussed in Andersen, Bollerslev, Diebold and Labys (2001a) (henceforth ABDL) as well as concurrent work by BarndorffNielsen and Shephard (2000, 2001)."

They compare their theoretical-based on covariance matrix method with actual VAR and GARCH popular methods.

Relevant information to include

The importance of realized volatility estimation. More cites to include: this one, Andersen, Bollerslev, Diebold and Labys (2001a) (henceforth ABDL) as well as concurrent work by BarndorffNielsen and Shephard (2000, 2001).

2.9 Corsi, Fulvio (2009) – A simple approximate long-memory model of realized volatility

Reference: [12] **Tags:** Keywords **Link:** https://doi.org/10.1093/jjfinec/nbp001

Problem statement and contribution

- Problem: More accurate and theory based model for forecasting Realized Volatility.
- Main contribution: Propose a novel cascade model, the HAR(3) that beat in interpretation and forecasting power to the standard AR-type models.
- Why it matters: It combines economic theory, specifically the heterogeneity market hypothesis, with a regression model to demonstrate several important results related to the realized volatility forecasting problem.

Commentaries

The definition of RV uses the square root.

Relevant information to include

Is the foundational paper of HAR.

2.10 Hansen, Peter R. and Lunde, Asger and Nason, James M. (2011) – The Model Confidence Set

Reference: [17] **Tags:** Model confidence set, model selection, forecasting, multiple comparisons **Link:** https://onlinelibrary.wiley.com/doi/abs/10.3982/ECTA5771

Relevant information to include

MCS procedure.

Introduces Model Confidence Set for selecting sets of superior models under loss-based comparison.

2.11 Andrew J. Patton (2011) – Volatility forecast comparison using imperfect volatility proxies

Reference: [29] **Tags:** Forecast evaluation, Forecast comparison, Loss functions, Realised variance, Range **Link:** https://www.sciencedirect.com/science/article/pii/S030440761000076X

Relevant information to include

Loss-function framework.

Establishes robust forecast evaluation principles; QLIKE is widely used in volatility forecasting comparisons.

2.12 Emmanuel Bacry and Iacopo Mastromatteo and Jean-Franccois Muzy (2015) – Hawkes processes in finance

Reference: [4] **Tags:** Keywords **Link:** Link

Commentaries.

Multivariate Hawkes applications overview.

Relevant information to include.

Positions Hawkes as workhorse for HFT finance problems including volatility estimation and order book modeling.

2.13 Du, Nan and Dai, Hanjun and Trivedi, Rakshit and Upadhyay, Utkarsh and Gomez-Rodriguez, Manuel and Song, Le (2016) – Recurrent Marked Temporal Point Processes: Embedding Event History to Vector

Reference: [14] **Tags:** stochastic process, recurrent neural network, marked temporal point process **Link:** https://doi.org/10.1145/2939672.2939875

Relevant information to include.

Neural TPPs.

Models event timing and marks jointly with RNN history embeddings.

2.14 Adam Clements and Yin Liao (2017) – Forecasting the variance of stock index returns using jumps and cojumps

Reference: [10] **Tags:** Realized variance, Jumps, Cojumps, Point process, Hawkes process, Forecasting **Link:** https://www.sciencedirect.com/science/article/pii/S0169207017300109

Problem statement and contribution.

- Problem: Use the intraday information available and modern jumps detection techniquess to test if HAR-based models can be improved by that way.
- Main contribution: Use jump indicators to enhance the performance of HAR based models at index-level.
- Why it matters: The variance forecasting is a principal issue in financial matters.

Commentaries.

Paper with very foundational information of jump detection.

Relevant information to include.

It states that [27] is the original source of HAR-RV, even tought, [12] was one of the applications. Rectification, the paper point out directly: "Among these forecasting models, the heterogeneous autoregressive (HAR) model proposed by Corsi (2009) has gained popularity due to its simplicity."

This paper attack the problem of using jumps and intraday information to enhance the HAR volatility forecast, therefore the new proposal should include the examination of a novel view, as including non-linear dynamics (LSTM).

Incorporating estimated jump intensity improves forecast accuracy, especially on jump/cojump days.

2.15 Hongyuan Mei and Jason Eisner (2017) – The Neural Hawkes Process: A Neurally Self-Modulating Multivariate Point Process

Reference: [10] **Tags:** None **Link:** https://arxiv.org/abs/1612.09328

Commentaries.

Can be useful in the next work that I am planning. Check how the proposed model is tested.

Relevant information to include.

Neural TPPs.

Learns complex, nonlinear dependence of intensity on event history beyond additive Hawkes kernels.

Basically built a continuous-time LSTM to modulate the intensity of the Hawkes.

2.16 Miura, Ryotaro and Pichl, Lukáš and Kaizoji, Taisei (2019) – Artificial Neural Networks for Realized Volatility Prediction in Cryptocurrency Time Series

Reference: [26] **Tags:** Keywords **Link:** https://link.springer.com/chapter/10.1007/978-3-030-22796-8_18

Relevant information to include.

In crypto realized volatility forecasting specifically, evidence is mixed: one study forecasting RV constructed from high-frequency Bitcoin returns compares ML models (including GRU/LSTM) to an optimized HAR-RV baseline and reports that a linear method (ridge regression) can outperform, reinforcing the strength of autoregressive long-memory structure in RV.

Reports ridge regression best among compared methods, supporting strong autoregressive RV dynamics in BTC.

2.17 Bucci, Andrea (2020) – Realized Volatility Forecasting with Neural Networks

Reference: [8] **Tags:** Volatility forecasting, Realized variance, HAR, HARQ, Robust regression, Weighted least squares, Box-Cox transformation, Forecast comparisons, QLIKE, MSE, VaR, Model confidence set **Link:** https://doi.org/10.1093/jjfinec/nbaa008

Relevant information to include

Reports recurrent networks (LSTM/NARX) outperform traditional econometric approaches for RV forecasting; notes interpretability trade-off.

2.18 Adam Clements and Daniel P.A. Preve (2021) – A Practical Guide to harnessing the HAR volatility model

Reference: [10] **Tags:** Volatility forecasting, Realized variance, HAR, HARQ, Robust regression, Weighted least squares, Box-Cox transformation, Forecast comparisons, QLIKE, MSE, VaR, Model confidence set **Link:** https://www.sciencedirect.com/science/article/pii/S0378426621002417

Problem statement and contribution

- Problem: Standard practices in HAR-RV modelling implies using raw RV and OLS estimators, which are far from optimal.
- Main contribution: Contribute with a survey of techniques to explore which methodology is more appropriate to harnessing HAR experiment. Its conclude that replacing OLS for WLS or RR typically leads to better adjustments, specially in longer horizon forecast.
- Why it matters: This relevant practices to implement HAR are valuables for anyone how want to applied the model.

Commentaries

Not clear if sqrt HAR or log HAR have better performance.

Relevant information to include

Several information have been found:

- Uses rolling window to explore the evolution of the β's.
- Using WLS or RR instead of OLS yields to better results, but the uses of square root or logarithm transformation of RV has not clear results.

2.19 Eunju Hwang and Won-Tak Hong (2021) – A multivariate HAR-RV model with heteroscedastic errors and its WLS estimation

Reference: [22] **Tags:** Multivariate HAR-RV model, GARCH(1,1) model, Weighted least squares estimation, Asymptotic normality **Link:** https://www.sciencedirect.com/science/article/pii/S0165176521001324

Problem statement and contribution

- Problem: Extending standard statistics tools to HAR model.
- Main contribution: Propose a multivariate HAR-GARCH framework with theoretical basis and fitted with WLSE.
- Why it matters: The RV has stylized features as long memory, non-gaussanity, and time varying volatility. HAR model the long memory, while GARCH handles the error non-normality assumption.

Commentaries

The innovation of the model is extending the methodology in the HAR-RV model while providing theoretical and simulation evidence before fitting it to real data.

Relevant information to include

A paper that uses WLS to get a better fit. It can be used as an argument for using WLS in the work.

2.20 Alla A. Petukhina and Raphael C. G. Reule and Wolfgang Karl Härdle (2021) – Rise of the machines? Intraday high-frequency trading patterns of cryptocurrencies

Reference: [31] **Tags:** Multivariate HAR-RV model, GARCH(1,1) model, Weighted least squares estimation, Asymptotic normality **Link:** https://www.tandfonline.com/doi/pdf/10.1080/1351847X.2020.1789684

Commentaries

Intraday HFT pattern analysis.

Relevant information to include

Documents intraday volatility periodicity and studies algorithmic-trading-like patterns in crypto.

2.21 Lykke Øverland Bergsli and Andrea Falk Lind and Peter Molnár and Michał Polasik (2022) – Forecasting volatility of Bitcoin

Reference: [35] **Tags:** Bitcoin, Volatility forecasting, Realized variance, HAR, GARCH **Link:** https://www.sciencedirect.com/science/article/pii/S0275531921001616

Problem statement and contribution

- Problem: Bitcoin Volatility.
- Main contribution: GARCH and HAR family models tested on this serie. $HAR > GARCH$
- Why it matters: Importance of a new currency with high volatility behavior.

Commentaries

HAR vs GARCH in bitcoin volatility forecast.

Relevant information to include

- HAR vs GARCH in bitcoin volatility forecast. HAR won.
- Realized variance with intraday data models outperform daily based methods, specially in shorter horizon.
- Importance of forecasting bitcoin volatility.

2.22 Matthias Kirchner and Silvan Vetter (2022) – Hawkes model specification for limit order books

Reference: [24] **Tags:** None **Link:** https://ideas.repec.org/a/taf/eurjfi/v28y2022i7p642-662.html

Relevant information to include.

Shows state-dependent baseline via imbalance; emphasizes careful specification; supports multivariate marked Hawkes for LOB dynamics.

2.23 Michael Mark and Jan Sila and Thomas A. Weber (2022) – Quantifying endogeneity of cryptocurrency markets

Reference: [25] **Tags:** None **Link:** https://doi.org/10.1080/1351847X.2020.1791925

Relevant information to include.

Reports Bitcoin dynamics largely endogenous; discusses 24/7 profile; notes kernel choice and long-memory issues.

2.24 Rafael Reisenhofer and Xandro Bayer and Nikolaus Hautsch (2022) – HARNet: A Convolutional Neural Network for Realized Volatility Forecasting

Reference: [32] **Tags:** None **Link:** https://arxiv.org/abs/2205.07719

Relevant information to include.

Aim to close the hap between NNs and HAR models, providing a called HARnet model that have better predictive accuracy on forecasting Bitcoin RV with tools to have interpretable parameters.

The interpretation shows that based on 1-week of history, the last two days are the most important, even when jumps occur. In the monthly horizon, it seems that the importance of the week information seems to diminish almost linearly when moving further into the past.

2.25 Raffaele Giuseppe Cestari and Filippo Barchi and Riccardo Busetto and Daniele Marazzina and Simone Formentin (2023) – Hawkes-based cryptocurrency forecasting via Limit Order Book data

Reference: [9] **Tags:** financial systems, time series forecasting, machine learning in finance, cryptocurrencies **Link:** https://arxiv.org/abs/2312.16190

Relevant information to include.

Shows Hawkes-rooted modeling of irregular LOB events can outperform benchmarks for return-sign prediction and profitability in simulations.

2.26 Lina Fan and Hao Yang and Jia Zhai and Xiaotao Zhang (2023) – Forecasting stock volatility during the stock market crash period: The role of Hawkes process

Reference: [16] **Tags:** HAR model, Realized volatility, Hawkes process, Forecast **Link:** https://doi.org/10.1016/j.frl.2023.103839

Problem statement and contribution.

- Problem: Can a Hawkes indicator enhance the HAR performance during crash period?
- Main contribution: Hawkes indicator can improve both in-sample and out-sample forecasting results.
- Why it matters: The inclusion of Hawkes process as a valuable regressor in HAR-RV model indicates the value of using microstructure to enhance macro (daily) feature forecast as RV.

Commentaries.

It is pretty similar of what I am proposing with a certain differences. The inclusion of the Hawkes is limited as a (not clear) regressor in the HAR-RV structure, this regressor is built under the exponential model, which is far from ideal with respect certain literature (I do not remember which exactly :c).

A step forward that I am proposing in the following article, is to not only use one type of Hawkes process, but further approaches and compare them also with LSTM, the specific hypothesis are presented at the beginning of the document.

Good paper to follow in order to have a structure and cites to use.

Relevant information to include.

Maybe the first attempt to combine Hawkes process with HAR-RV model to improve the forecast.

Hawkes indicator improves volatility forecasting during crash period; predictive for many stocks

2.27 Anupam Dutta and Elie Bouri (2024) – Forecasting the volatility of crude oil futures: New evidence from jump-induced volatility

Reference: [15] **Tags:** Crude oil futures, Realized volatility, Jump-induced volatility, OVX, Leverage effects, HAR-RV models **Link:** https://www.sciencedirect.com/science/article/pii/S2211467X24002979

Problem statement and contribution.

- Problem: Crude Oil volatility forecast using time-varying jumps.
- Main contribution: "Jumps offer added information for forecasting the RV of crude oil futures"
- Why it matters: Add information of crude oil problem.

Commentaries.

Relevant information to include.

[15] uses GARCH-jump approach to obtain jump detection and intensity for enhancing the HAR model using them as regressors.

2.28 Saef, Danial and Nagy, Odett and Sizov, Sergej and Härdle, Wolfgang Karl (2024) – Understanding temporal dynamics of jumps in cryptocurrency markets: evidence from tick-by-tick data

Reference: [33] **Tags:** Keywords **Link:** https://doi.org/10.1007/s42521-024-00116-1

Problem statement and contribution.

- Problem: How the state of art of jump detection and high frequency analysis applies to Cryptocurrencies (CCs)
- Main contribution: Uses of jump detection literature on CCs data in a extensive research.
- Why it matters: New market, maybe new rules, needs to be examinated.

Commentaries.

High range of data and examples.

Inclusion of important world events in time to compare with CCs.

Relevant information to include.

For crypto specifically, the empirical microstructure literature documents (i) 24/7 trading, (ii) strong intraday periodicity that still aligns with global risk-hours, (iii) frequent jumps and clustering, and (iv) strong cross-venue linkages—precisely the setting where "shock-aware" intensity signals are plausible incremental predictors.

Jumps on up to 58% of days, negative dominance, systematic clustering; strong intraday/time-zone patterns; highlights need for noise-robust tools and multi-asset cojump methods

2.29 Lei Zhang and Yan Chen and Elie Bouri (2024) – Time-varying jump intensity and volatility forecasting of crude oil returns

Reference: [34] **Tags:** Crude oil market, Volatility forecasting, GARCH-based model, Time-varying jump intensity **Link:** https://www.sciencedirect.com/science/article/pii/S014098832300734X

Problem statement and contribution

- Problem:
- Main contribution:
- Why it matters:

Commentaries

Relevant information to include

Hawkes jump intensity improves volatility forecasting, especially during COVID.

2.30 Genhua Hu and Xiaoqing Ma and Tingting Zhu (2025) – A Forecasting volatility of China's crude oil futures based on hybrid ML-HAR-RV models

Reference: [21] **Tags:** China's crude oil futures, High-frequency data, Signed jumps, HAR-RV model, Machine learning **Link:** https://www.sciencedirect.com/science/article/pii/S1062940825000683

Relevant information to include

HAR variants and ML hybridization.

2.31 Sohaib El Karmi (2025) – A Deterministic Limit Order Book Simulator with Hawkes-Driven Order Flow

Reference: [23] **Tags:** None **Link:** https://arxiv.org/abs/2510.08085

Relevant information to include

Multivariate marked Hawkes; compares exponential vs power-law; GOF via time-rescaling; nearly unstable regime emphasized

Hawkes-driven LOB simulator

Highlights importance of nearly-unstable regime for realistic clustering; provides reproducible code and artifacts

2.32 Alexander Brauneis and Mehmet Sahiner (2026) – Crypto Volatility Forecasting: Mounting a HAR, Sentiment, and Machine Learning Horserace

Reference: [7] **Tags:** Cryptocurrencies; Sentiment; Machine learning; Volatility forecasting **Link:** https://ideas.repec.org/a/kap/apfinm/v33y2026i1d10.1007_s10690-024-09510-6.html

Problem statement and contribution

- Problem: Sentiment in HAR prediction.
- Main contribution: Test HAR and many ML models with and without a sentiment variable. The ML increases their performance while HAR doesn't. Sentiment act in a non-linear way.
- Why it matters: Other features may not interact in a linear way with the variable that we want to predict, therefore non-linear methods are necessary.

Commentaries

Very clear paper to follow in case to replicate.

Relevant information to include

HAR and ML in crypto volatility forecast. Inclusion of other variables may include new information for increasing forecast performance but in non-linear ways, for instance, the sentiment.

ML improves accuracy vs HAR; argues certain predictors influence volatility nonlinearly, captured by ML but not HAR; sentiment improves ML forecasts in 54.17% of cases studied

3 General notes

3.1 Project Viability

This subsection outlines the minimal elements required before proceeding with further development. In particular, it clarifies the research question, situates it within the existing literature, defines the proposed methodology, and highlights the expected contribution. The goal is to ensure a coherent and empirically identifiable research design.

Research Question

Can intraday event-driven information—ranging from simple extreme-event counts to Hawkes-based intensity measures—improve realized volatility forecasting in cryptocurrency markets, and does the predictive value of such information depend on the model class used to incorporate it?

Brief View of the State of the Art

Forecasting realized volatility (RV) is a central problem in financial econometrics due to its relevance for risk management, derivative pricing, and portfolio allocation [1, 2, 3].

The Heterogeneous Autoregressive (HAR) model [12] provides a simple and effective framework to capture long-memory dynamics in volatility through daily, weekly, and monthly components.

This structure is motivated by the Heterogeneous Market Hypothesis [27], which posits that market participants operate at different time horizons, thereby generating persistence across multiple temporal scales.

Despite its strong empirical performance and interpretability, HAR relies exclusively on past realized volatility and ignores information contained in intraday market activity. Several extensions have incorporated additional features such as jump detection [5], semivariance decomposition [6, 30], and other exogenous variables (see, e.g., [21]). While these approaches improve forecasting performance, they remain largely within a linear modeling framework.

A commonly studied class of covariates in volatility forecasting is based on jump detection and jump intensity [10, 15, 34, 21]. These features often improve predictive accuracy; however, their integration is typically linear, which may not fully capture their informational content.

Recent evidence suggests that nonlinear models may better exploit complex information. For instance, [7] shows that sentiment-based variables improve volatility forecasting more effectively in machine learning models than in HAR-type models. This highlights the potential limitations of linear specifications when incorporating rich, nonlinear features.

Cryptocurrency markets, in particular, exhibit strong endogenous dynamics and clustering of events [31, 25, 33]. Such behavior is naturally modeled by Hawkes processes [18], which capture self-exciting dynamics in event arrivals. Their application has been explored in cryptocurrency settings, including simulation [24], forecasting [9], and stress-period analysis [16].

Despite these advances, it remains unclear whether event-based features—particularly those derived from Hawkes processes—consistently improve volatility forecasting, and whether their effectiveness depends on the modeling framework used to process them. This gap motivates a structured comparison between linear and nonlinear models under progressively richer information sets.

Methods and Experiments

The empirical strategy is designed to isolate the incremental value of event-driven information and its interaction with model class.

First, intraday data at a 5-minute frequency are collected for a set of cryptocurrencies (BTC, ETH, SOL, XRP, ADA) from Binance over a multi-year period. This frequency represents a standard compromise between noise reduction and information retention in high-frequency volatility estimation [2, 11]. These data are used to construct daily realized volatility series, with a logarithmic transformation applied for stability.

Second, extreme intraday events are identified using a dynamic threshold based on conditional quantiles, following [19]. An event is defined whenever the absolute return exceeds the threshold, and the daily count of such events, denoted N_t, is computed. This step provides a simple, model-free representation of event-driven activity.

Third, Hawkes processes are fitted to the event times using exponential kernels. The conditional intensity function is given by

$$\lambda(t) = \mu + \sum_{t_i < t} \alpha e^{-\beta(t-t_i)}, \tag{1}$$

and the daily integrated intensity

$$\Lambda_t = \int_{\text{day } t} \lambda(s)\, ds \tag{2}$$

is used as a feature capturing endogenous clustering effects. This provides a model-based representation of event dynamics.

Fourth, forecasting experiments are conducted using both HAR-RV and LSTM models. HAR-RV is estimated via OLS as the main benchmark (and alternatively WLS [11] as a robustness check), while LSTM models are trained using AdamW optimization. All models are evaluated on the same information sets to ensure comparability.

Fifth, models are evaluated along an information ladder:

$$[RV] \;\rightarrow\; [RV, N_t] \;\rightarrow\; [RV, \Lambda_t],$$

which allows us to identify: (i) the value of event detection, (ii) the additional value of Hawkes-based modeling, and (iii) whether nonlinear models better exploit these features.

Finally, model performance is evaluated using QLIKE loss [29], along with MSE and MAE. Statistical inference is conducted using the Model Confidence Set (MCS) procedure [17] and the Diebold–Mariano (DM) test [13].

Robustness Checks

To ensure the validity and generality of the results, several robustness analyses are conducted.

Event Definition

- alternative quantile thresholds,
- fixed-threshold specifications,
- sensitivity to threshold selection.

Hawkes Specification

- exponential kernel (baseline),
- power-law and multi-exponential kernels,
- alternative estimation procedures.

Feature Construction

- alternative aggregation of intensity (mean vs integrated),
- inclusion/exclusion of event counts and intensity jointly.

Model Specification

- HAR estimated via OLS (baseline) and WLS,
- different LSTM architectures (depth, hidden dimension),
- alternative input representations.

Forecast Design

- alternative forecast horizons (1-day, 7-day, 30-day),
- rolling window configurations.

Cross-Asset Validation

- primary assets: BTC, ETH,
- robustness assets: ADA, XRP, SOL.

Expected Outcome and Contribution

The central hypothesis is that event-driven features contain additional predictive information for realized volatility. More precisely, the contribution of such information is evaluated through a structured empirical ladder and across different model classes.

The contribution of this work is threefold. First, it introduces a systematic comparison between simple event counts and Hawkes-based intensity measures as predictors. Second, it evaluates whether the predictive value of these features depends on the modeling framework. Third, it provides empirical evidence on the interaction between event-driven information and model nonlinearity in cryptocurrency markets.

More broadly, the results aim to clarify whether improvements in volatility forecasting arise from richer features, more flexible models, or the interaction between both, thereby contributing to the integration of econometric and machine learning approaches in financial modeling.

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